

CSCI 1101 Introduction to Computer Science



WHAT IS COMPUTER SCIENCE
AND
A [BRIEF] HISTORY OF COMPUTERS AND
COMPUTING

What is computer science?



Computer Science is the study of computer systems including hardware and software.

Unlike computer engineers who deal with hardware and software, computer scientists focus mainly on software (programs) and software systems (which includes their theory, design, development, and application). This requires them to solve from abstract problems (like determining if a problem can be solved with a computer and, if it is, discovering the algorithm that will solve it) to concrete problems like creating a web site, installing and maintaining a network, designing and implementing a device driver, etc.

As a CS student you will learn not only programming languages, but how to design computer systems, how people interact with computers, how to handle large amounts of information, how to build networks, create websites, computer animation, robotics, and much more.

Computer Science



As a discipline

Discipline:

- A field of study.
- A branch of knowledge, typically one studied in higher education.

- What can be efficiently automated?
- Necessary skills
 - Design solutions
 - Algorithmic thinking
 - Data Representation
 - Programming

Terminology

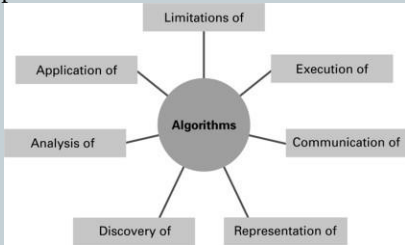


- **Algorithm:** A set of steps that defines how a task is performed
- **Program:** A representation of an algorithm
- **Programming:** The process of developing a program
- **Hardware:** Equipment; the physical elements of the system
- **Software:** Programs and algorithms
- **Abstraction:** a mental model that removes complex details

The Role of Algorithms in CS



- Algorithms play a central role in the discipline of computer science



Systems Areas and Application Areas

Systems: Helps manage computer system

- Algorithms and Data Structures
- Programming Languages
- Architecture
- Operating Systems
- Software Engineering
- Human-Computer Communication

Applications: Get stuff done

- Numerical and Symbolic Computation
- Databases and Information Retrieval
- Intelligent Systems
- Graphics and Visual Computing
- Net-Centric Computing
- Computational Science

History of the Computer



- <http://www.history.com/shows/modern-marvels/videos/who-invented-the-computer>
- <http://www.computerhistory.org/timeline/?category=computer>

Early (not electronic) Computing Machines



- **Abacus:** positions of beads represent numbers
- **Gear-based machines:** positions of gears represent numbers
 - Blaise Pascal: Pascaline, mechanical calculator
 - Joseph Jacquard: Jacquard's Loom, inventor of punch card
 - Charles Babbage: Difference Engine, Analytic Engine
 - Ada Lovelace: first programmer, inventor of the loop

The “Generations” of Computing



- Hardware and software innovations went hand in hand
- Each “generation” is characterized by particular technology

First Generation (1951-1959)

Hardware

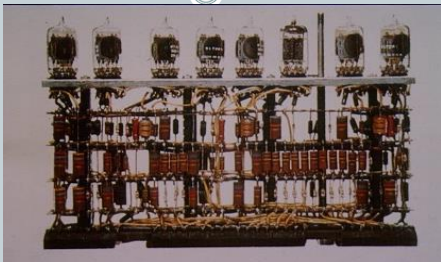
- Mechanical relays (Harvard Mark I)
- Vacuum tube (Colossus, ENIAC)
- Magnetic drum
- Magnetic tape drive



Software

- Earliest machine programs were hard wired
- Machine language
- Assembly language
- Programmers already splitting into systems and applications

First Generation

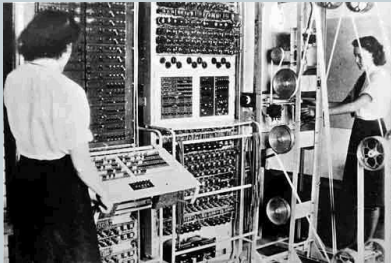


<http://minf.vub.ac.be/~marc/info-hecol/html/cu-02.html>

First Generation



- Colossus, United Kingdom, 1944



Second Generation (1960-1965)



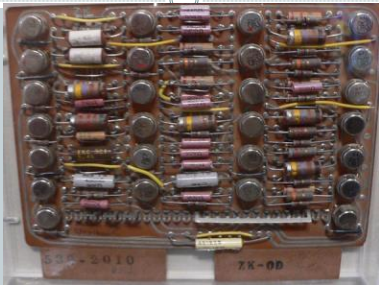
Hardware

- Transistors
- Magnetic core memory
- Magnetic discs

Software

- Introduction of high-level programming languages

Second Generation

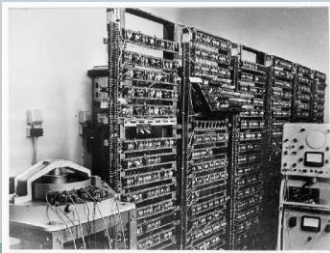


<http://www.ganssle.com/tem/tem257.html>

Second Generation



- Manchester University Experimental Transistor Computer, 1953



Third Generation (1965-1970)

Hardware

- Integrated circuits
- Transistors for memory
- Terminal

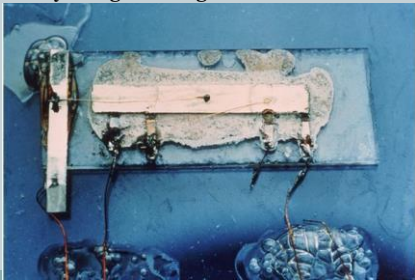
Software

- Operating system
- Increasing separation between user and hardware

Third Generation



- Jack Kilby's original integrated circuit



Third Generation



Cromemco, "ZPU" (4mHz Zilog Z80 CPU board) 1976

<http://www.kuhmann.com/Cromemco/CrosNest.htm>

Fourth Generation (1970 – 1990)



Hardware

- Advances in chip technology
- Miniaturization and embedded circuits
- Emergence of the personal computer
- Introduction of networking

Software

- Proliferation of application software
- Rapid growth of computer use

Fourth Generation



<http://www.directindustry.com/prod/eurotech/cpu-boards-7026-846807.html>

Fourth Generation

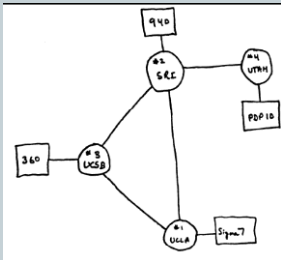


<http://as400andmainframefiles.wordpress.com/2013/01/24/2005-gif.jpg>

Fourth Generation

- **ARPANET: The beginnings of the Internet**

- Original: 5 nodes
- Now: 500+ million nodes



Fifth Generation (1990-present)



Hardware

- Continued LSI and miniaturization
- Mobile computing

Software

- Networking
 - Internet
 - World Wide Web
- Object oriented design
- Explosion of number of users

Fifth Generation



<http://www.technews.net/2009/11/05/garmin-aera-series-handheld-navigation-devices/>

<http://hgr.com/tag/tablets/>

<http://www.learnsmartsystems.com/details/mobile.asp>

x

